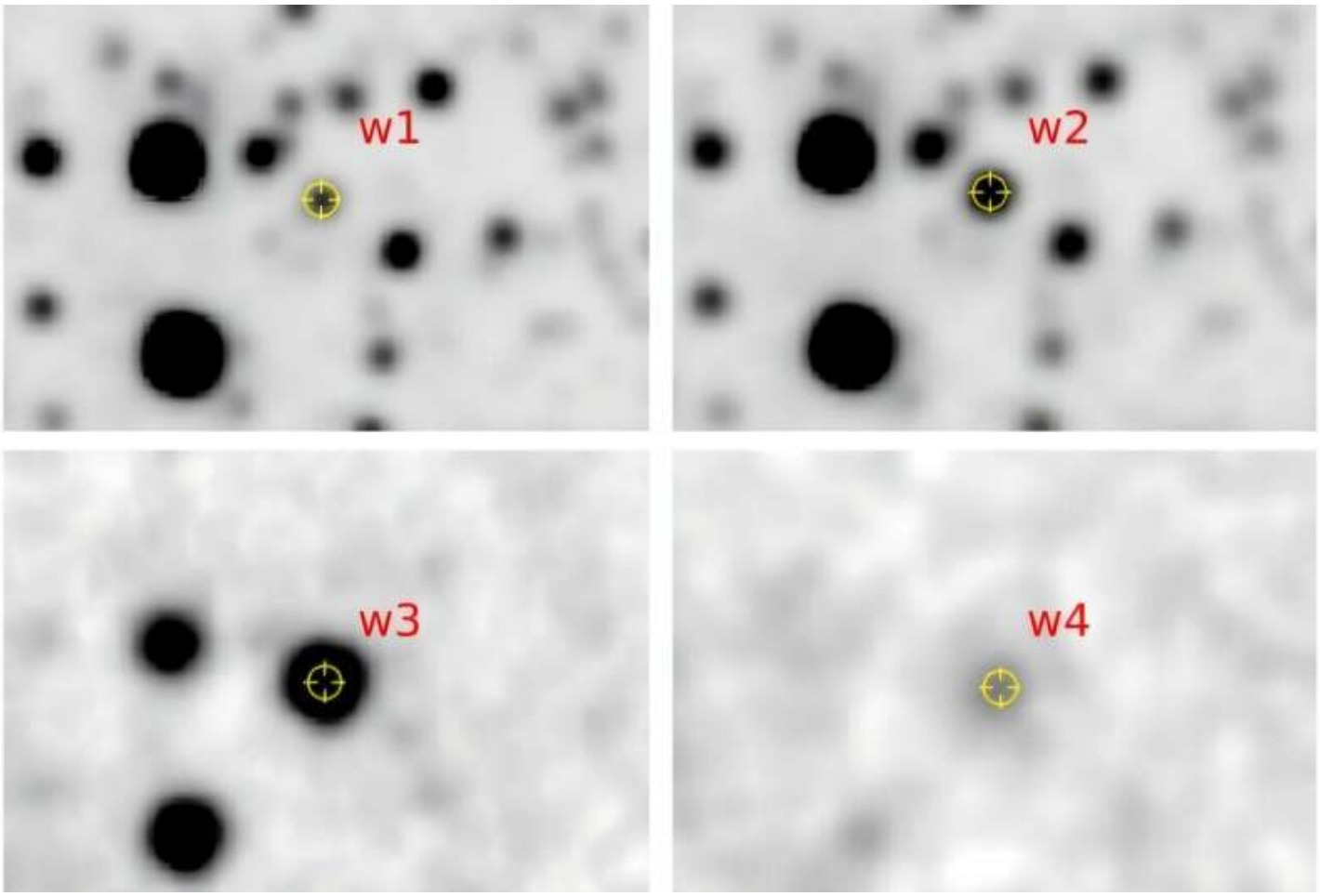
Disentangling a dusty nova: Astronomers take a closer look at LMCN 2009-05a

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A mosaic of a 3x3 arc minute square field around LMCN 2009-05a. The source is detected in all 4 WISE bands: W1 (3.4 μm), W2 (4.6 μm), W3 (12 μm) and W4 (22 μm); the emission at W2 and W3 bands is pronounced. Credit: arXiv (2025). DOI: 10.48550/arxiv.2508.13851

By analyzing the data from the American Association of Variable Star Observers (AAVSO) international database and the Small and Medium Aperture Telescope System (SMARTS), astronomers have inspected a nova in the Large Magellanic Cloud (LMC) known as LMCN 2009-05a. Results of the study, published August 19 on the pre-print server *arXiv*, disentangle the nature and properties of this nova.

A nova is a star that suddenly gets much brighter and then slowly returns to its original state, over a period of several months. Such a powerful outburst is the result of the accretion process in a close binary system containing a white dwarf and its companion. Studying novae can therefore help us

better understand fundamental astrophysical processes, including stellar evolution.

Novae are also good places to inspect dust formation, which may occur during outbursts within a few weeks to a few months after the eruption.

LMCN 2009-05a, which was first reported in May 2009, is one such nova exhibiting dust formation and a team of astronomers led by Mohit Singh Bisht of the Indian Center for Space Physics in India decided to take a closer look at this eruption in order to shed more light on its properties.

According to the new study, AAVSO and SMARTS data indicate that LMCN 2009-05a was a moderately fast nova, as its decline by two magnitudes from the maximum light took only 46 days. The absolute magnitude of LMCN 2009-05a was measured to be -6.65, which shows that it was a relatively low luminosity nova with an outburst luminosity at a level of 46,400 solar luminosities.

The optical and near-infrared light curves suggest that a significant amount of dust formed in the ejecta of LMCN 2009-05a between 78 and 155 days after the outburst. The dust temperature, measured on day 395 after the outburst, turned out to be approximately 700 K.

By analyzing the spectroscopic observations of LMCN 2009-05a, Bisht's team found that the spectra are initially dominated by hydrogen Balmer and iron lines with P-Cygni profiles. All in all, the results indicate nitrogen and oxygen abundances significantly enhanced relative to solar values.

Furthermore, the study found that the white dwarf progenitor of LMCN 2009-05a had a relatively low mass—of around 0.77 solar masses. Based on the collected data, the astronomers also calculated that the expansion velocity of the ejecta was approximately 690 km/s.

Summing up the results, the authors of the paper made some remarks regarding the overall nature of LMCN 2009-05a.

"The spectrophotometric evolution of LMCN 2009-05a shows that it is a classical moderately fast nova. This nova is marked by notable dust formation and changes in ionization structure. It fits the D-class light curve morphology," the researchers conclude.

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More information: Mohit Singh Bisht et al, The Unexplored Dusty Nova LMCN 2009-05a in the Large Magellanic Cloud, *arXiv* (2025). DOI: 10.48550/arxiv.2508.13851

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